
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Although Mixture of Experts (MoE) models have become indispensable to the scaling of large language models, the mechanistic respects in which they diverge from dense networks remain conspicuously underexplored. A substantial body of prior work has characterized how dense models exploit *superposition* to encode a greater number of features than they possess dimensions, and has established that the extent of such superposition is governed jointly by feature sparsity and feature importance; MoE models, however, resist mechanistic explanation through this same interpretive lens. We demonstrate that neither feature sparsity nor feature importance precipitates the discontinuous phase changes characteristic of dense networks, and that network sparsity—the ratio of active to total experts—constitutes a more faithful characterization of MoE behavior. To this end, we introduce novel metrics for quantifying superposition across experts, and our analysis establishes that models exhibiting greater network sparsity concomitantly display greater *monosemanticity*. Departing from conventional load-balancing accounts, we advance a reconceptualized definition of expert specialization grounded in monosemantic feature representation, and we show that experts, when suitably initialized, spontaneously organize themselves around coherent feature combinations. Taken together, these results intimate that network sparsity in MoEs may afford more interpretable models without any attendant sacrifice in performance, thereby contesting the prevailing supposition that interpretability and capability stand in fundamental opposition.

1 INTRODUCTION

Mixture of Experts (MoEs) have attained pervasive adoption within state-of-the-art language models—among them Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025)—owing chiefly to the computational efficiency and performance gains they confer (Shazeer et al., 2017; Fedus et al., 2022). A succession of subsequent contributions has refined the routing apparatus (as exemplified by Expert-Choice routing) and fortified training stability and transfer (as with ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical findings alike further attest that learnable routers are capable of discovering latent cluster structure within data, thereby furnishing an account of why experts come to specialize (Dikkala et al., 2023). Notwithstanding their widespread deployment, however, MoEs remain imperfectly understood when scrutinized through the prism of mechanistic interpretability.

Interpretability-oriented approaches have endeavored to render expert behavior more transparent. Yang et al. (2025b) advance MOE-X, which promotes sparsity-aware routing and enlists wide, ReLU-based experts so as to attenuate polysemanticity, whereas Park et al. (2025) introduce MONET, wherein the proliferation of experts is exploited to enable capability editing through expert activation. These works, however, remain preoccupied with architectural modification; a mechanistic understanding of how MoEs represent features, of how experts modulate superposition, and of whether experts specialize spontaneously in the absence of supplementary regularization continues to elude us. In their survey of the MoE

literature, Mu & Lin (2025) accordingly single out mechanistic interpretability as a pivotal open challenge.

A foundational obstacle confronting the interpretation of neural networks is the phenomenon of superposition—the circumstance in which models represent a greater number of features than the dimensions available to them. Such a regime permits networks to compress numerous sparse features into a smaller complement of neurons, though only at the expense of rendering individual neurons polysemantic and, in consequence, refractory to interpretation.

MoE architectures introduce an additional axis to this problem, namely network sparsity. Whereas dense models activate the entirety of their neurons irrespective of the input presented, MoEs engage only a fraction of their total parameters (Shazeer et al., 2017). Dense models thus exploit feature sparsity by packing a multitude of sparse features into shared neurons; MoEs, by contrast, can afford a greater degree of selectivity, potentially consecrating entire experts to particular feature combinations.

We accordingly interrogate whether (1) MoEs manifest less superposition than their dense counterparts, (2) there obtains a discrete phase change in the quantum of superposition allotted to a particular feature within MoE experts as a function of its relative importance and of overall feature sparsity, as is observed in dense models, and (3) expert specialization admits of interpretation through the lens of feature representation rather than through load balancing alone.

We pursue these questions by means of simple models that extend the framework of Elhage et al. (2022) to the MoE setting. Our principal contributions are as follows: (1) in contradistinction to dense models, MoEs do not exhibit sharp phase changes, evincing instead more continuous transitions as network sparsity is increased; (2) MoEs consistently display greater monosemanticity—that is, less superposition—than dense models endowed with equivalent active and total parameters, with individual experts representing features more cleanly; and (3) we advance an interpretability-focused definition of expert specialization predicated upon monosemantic feature representation, thereby demonstrating that experts organize themselves spontaneously around coherent feature combinations rather than around arbitrary load balancing.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis posits that networks encode concepts as directions within activation space (Park et al., 2024), and yet the number of interpretable features routinely surpasses the dimensions on hand. Elhage et al. (2022) formalized this phenomenon under the rubric of superposition, adducing evidence that dense models rely upon non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. Although superposition is theoretically efficient (Scherlis et al., 2025), its presence necessitates elaborate post-hoc disentanglement procedures—Sparse Autoencoders foremost among them—in order to recover monosemantic features (Bricken et al., 2023). A recent current of work has interrogated how data correlations modulate superposition (Prieto et al., 2025), together with the manner in which interference patterns arise and may be attenuated (Gurnee et al., 2023). Our contribution prolongs this line of inquiry into MoE architectures, establishing that network sparsity—rather than feature sparsity in isolation—is what governs representational strategy.

Interpretability of MoEs. Although MoEs have emerged as the canonical instrument for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), a mechanistic understanding of their internal representations lags conspicuously behind that of their dense counterparts. Extant analysis remains preoccupied for the most part with macroscopic behaviors—among them routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). More recent, interpretability-oriented approaches endeavor to compel specialization by way of architectural constraints, whether through predefined concept routing (Yang et al., 2025b) or through the scaling of expert counts so as to match vocabulary sizes (Park et al., 2025); such approaches, however, tend to privilege capability editing over any explanation of intrinsic

feature geometry. We redress this lacuna by analyzing how experts modulate superposition, demonstrating thereby that MoEs exhibit greater monosemanticity than dense models and proposing a feature-based definition of expert specialization.

3 BACKGROUND

A principal preoccupation of Mechanistic Interpretability is the reverse engineering of neural networks; one such methodology consists in decomposing model representations into a repertoire of human-interpretable concepts designated ‘features’. These features are frequently presumed to be linear, which is to say that any hidden state h admits of the description

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

wherein $\vec{f}_i$ denotes the direction corresponding to feature i , α_i denotes the activation strength of that feature (roughly, the degree to which feature i is present within the input), and F denotes the set of all represented features.

The *superposition hypothesis* maintains that models are capable of representing far more features than dimensions, that is, $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). For a latent space to accommodate many more features than dimensions, the feature vectors comprising F must be packed in such a manner that they are not all mutually orthogonal. Whenever two features exhibit interference $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say that they reside in *superposition*. Such superposition is tolerable insofar as features are only sparsely active—a property characteristic of the preponderance of online text data—yet it is procured at the cost of interpretability, since α_i thereby harbors spurious activations unrelated to the presence of feature i within the input.

Monosemanticity constitutes a property of individual neurons, whereby a neuron’s activation corresponds cleanly to a single α_i —that is, features are basis-aligned. Should features be orthogonal (and hence not in superposition) yet fail to be basis-aligned, neurons persist in being *polysemantic* notwithstanding the minimality of feature interference. Throughout this paper, our emphasis falls upon the reduction of superposition rather than upon the enforcement of basis-alignment. For the sake of brevity, whenever we characterize models, experts, or features as “more monosemantic,” we intend thereby that they display less superposition.

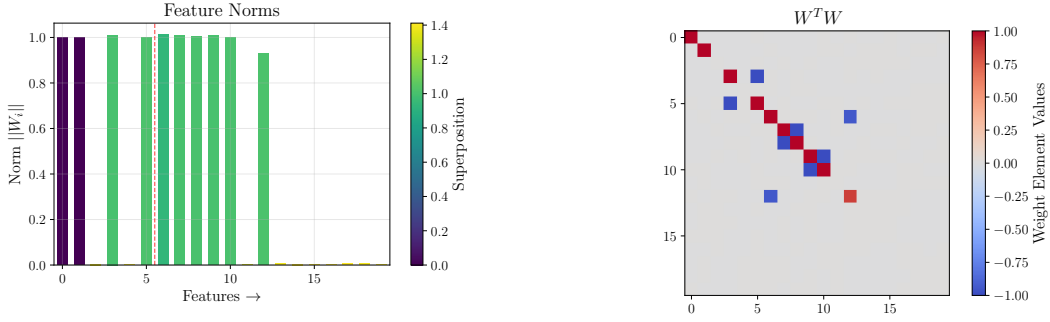
4 DEMONSTRATING SUPERPOSITION

MoEs are frequently conceptualized as compositions of dense models, wherein each expert is presumed to behave in the manner of an independent dense network. Whether experts in fact represent features in a fashion analogous to dense models nonetheless remains an open question. We adjudicated this matter by comparing the respects in which MoEs and dense models differ with regard to superposition.

4.1 EXPERIMENTAL SETUP

Our objective is to elucidate the manner in which a MoE may project a high-dimensional vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and thereafter recover it with fidelity. This undertaking extends the framework of Elhage et al. (2022) to the MoE setting.

The input distribution. The input vector x represents the activations of an idealized, disentangled model, wherein each dimension x_i corresponds to a distinct and independent feature—in effect, a perfectly neuron-aligned and monosemantic representation. We take $x_i \sim U(0, 1)$, save that, for a prescribed sparsity $S \in [0, 1)$, we impose $P(x_i = 0) = S$. Inasmuch as features do not all contribute equally to the loss, and in order to model this heterogeneity of utility, we assign to each feature x_i a scalar importance I_i that weights the reconstruction loss. Additionally, so as to isolate and interrogate the phase transitions



(a) Norm of each feature’s weight vector $\|W_i\|$, wherein color encodes superposition status—green designating features residing in superposition and purple designating monosemantic features.

(b) $W^T W$ matrix in which each cell represents $(\hat{W}_i \cdot W_j)$, thereby laying bare the interference patterns obtaining between features.

Figure 1: Feature representation and superposition within a dense model comprising $n = 20$ features and $m = 6$ hidden dimensions, under importance $I = 0.7^i$ and a uniform feature density of $(1 - S) = 0.1$. Superposition (rendered as color) is given by $\sum_j (\hat{W}_i \cdot W_j)^2$.

pertaining to a single feature, we scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, such that $I = (1, 1, \dots, r)$. In this manner, I_i and r jointly modulate the signal strength of the features, thereby enabling us to probe the sensitivity of experts to feature magnitude.

Model Architecture. The MoE comprises E experts, each expert e being parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ together with a bias $b^e \in \mathbb{R}^n$. Inputs are consigned to the top- k experts by way of a learned router $g(x) = \text{softmax}(W^r x)$ in which $W^r \in \mathbb{R}^{E \times n}$. Each active expert projects the input onto a lower-dimensional hidden state $h^e = W^e x$ and thereupon generates a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^T h^e + b^e)$ ¹. The final output constitutes the weighted sum of the active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, in which the w_e are the renormalized gating weights.

We train our models under an L2 reconstruction loss weighted by the feature importances I_i , namely $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, so as to forestall expert collapse, we adjoin the standard auxiliary load-balancing loss (Fedus et al., 2022), defined by $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Herein N denotes the total number of experts, f_e the fraction of samples within a batch routed to expert e , P_e the average gating probability apportioned to expert e across the batch, and $\alpha = 0.01$ the coefficient governing the strength of the penalty. We deliberately dispense with auxiliary load balancing in Section 4 in order to isolate the intrinsic architectural bias of the MoE with respect to superposition.

4.2 MEASURING FEATURE CAPACITY

In order to analyze feature representations across architectures, we compared two fundamental properties of the features, namely their *representation strength* and their *interference* with other features. We measured the norm of a feature’s weight vector within an expert e , given by $\|W_i^e\|$, which quantifies the extent to which a feature is represented within that expert: $\|W_i^e\| \approx 1$ when feature i is fully represented within expert e , and zero when it goes unlearned. We render visible the interference of a feature i with other features within expert e by means of the Gram matrix $W^T W$, whose off-diagonal elements encode pairwise interference.

¹The output ReLU guarantees non-negative reconstructions, in consonance with our input distribution, wherein $x_i \in [0, 1]$ whenever nonzero. Moreover, the off-diagonal terms of $(W^e)^T W^e$ engender negative interference, which the ReLU proceeds to suppress; when features are sparse, such negative interference becomes effectively “free,” being filtered to zero, and thereby incentivizes model configurations possessing negative off-diagonal terms (antipodal pairs being a case in point).

As is evident from Figures 1a and 2a, the dense model and the MoE represent a comparable number of features (10 as against 8) with commensurate norms for an equal budget of total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$). Although the MoE experts do exhibit a measure of local superposition (Expert 0 in Fig 2b being illustrative), the global interference structure remains strictly partitioned. Whereas in the dense model (Figure 1b) any feature is free to interfere with any other, the MoE imposes a block-diagonal structure in which features routed to distinct experts incur zero interference. This attests that MoEs allocate representational capacity by partitioning the feature space, thereby curtailing the global scope of interference.

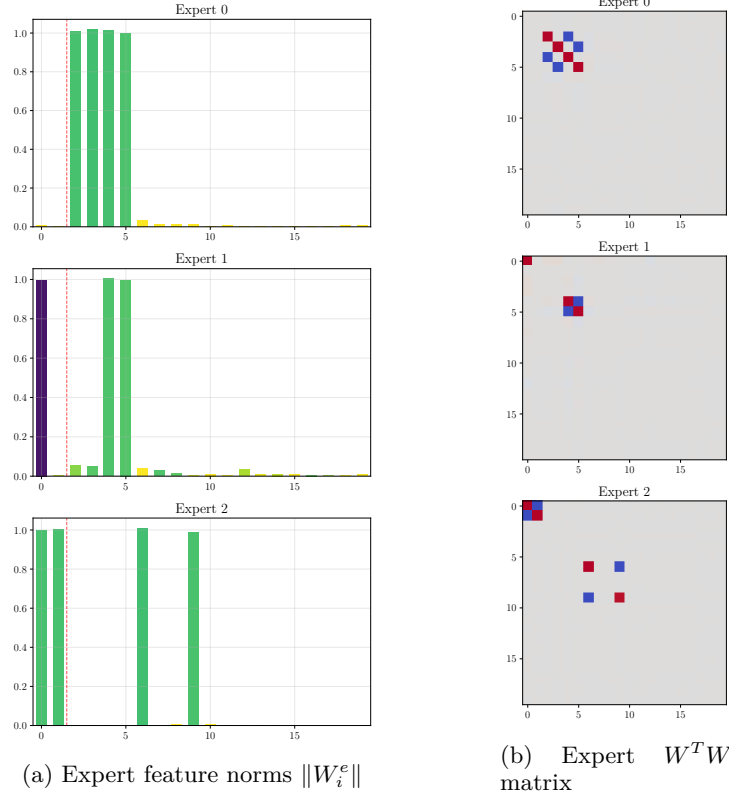


Figure 2: Feature representation and superposition within a MoE comprising $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (under top- $k = 1$ routing), with importance $I = 0.7^i$ and a feature density of $1 - S = 0.1$.

Expert Feature Dimensionality. We sought to apprehend how MoEs allocate their circumscribed representational capacity in a manner divergent from that of the dense model. To this end we measured feature dimensionality, which quantifies the “fraction of a dimension” apportioned to a specific feature within a model (Elhage et al., 2022). For a feature i , we define its dimensionality within expert e by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero (unlearned) and one (monosemantic). The aggregate capacity of a MoE may accordingly be defined as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. Whenever features are “efficiently packed” within a model’s representation space, the dimensionalities of the several features sum to the number of embedding dimensions, that is, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al.,

2022). In the case of a MoE, this relation generalizes to $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, we find that dense and MoE models alike satisfy the foregoing dimensionality constraint, which is to say that MoEs attain the same efficiency in packing features as do dense models for an equivalent budget of total parameters.

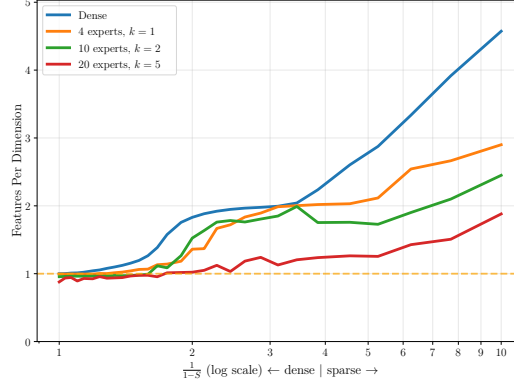


Figure 3: Features per dimension plotted against inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) harbors the most superposition, a quantity that diminishes monotonically as the expert count is augmented: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). All models possess equal total parameters and comparable k/E . The dashed line at 1.0 demarcates monosemantic representation.

Features per Dimension & Network Sparsity. Inasmuch as dense and MoE models alike “efficiently pack” features within their representation space, we compared the disparities in the number of features per dimension across the two model classes. This comparison afforded us an exact measure of superposition in both models, together with a means of discerning how the number of experts in a MoE bears upon superposition across differing feature sparsities. Should the features per dimension exceed unity, the features necessarily reside in superposition, the model representing more features than it possesses dimensions. We define features per dimension for a MoE by

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ denotes the Frobenius norm and p_e denotes the expected probability that expert e is invoked across a batch of input samples—that is, the average renormalized gating weight consequent upon top- k routing.

For MoEs endowed with an *equal* number of total parameters as the dense model, we observe that the dense model registers a higher number of features per dimension (Figure 3) and hence more superposition. This observation indicates that the dense model harnesses superposition to represent a greater total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE inclines toward capping its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Furthermore, as we augment the total number of experts within the MoE—while holding the total parameters and the ratio k/E approximately fixed—the number of features per dimension declines, which is to say that superposition abates. *The greater the number of experts, the less superposition the model exhibits.* Concretely, features grow more monosemantic as the number of experts increases. It bears noting, moreover, that the surfeit of superposition in the dense model enables it to attain consistently lower reconstruction loss than the MoEs, as displayed in Figure 6 of Appendix A.1, the loss differential at any given sparsity amounting to ~ 0.03 – 0.08 . Yet as the number of experts increases, the MoEs come

to attain loss consistently comparable to that of the dense models. A theoretical intuition is furnished in Appendix A.2.

5 PHASE CHANGE

Although MoEs and dense models learn a comparable number of features, MoEs distribute those features across experts with diminished interference. This circumstance intimates that network sparsity reconfigures the manner in which features are allocated rather than the number that are learned. We examined how properties of the input distribution—feature sparsity and importance among them—orchestrate this allocation, and whether they induce the physics-inspired *phase changes* in representation.

Models possess only a finite repertoire for representing features: each feature may be ignored, superimposed, or rendered monosemantic. Phase change designates the observation that discrete boundaries occasionally demarcate these regions, boundaries that are themselves functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). By varying the sparsity and relative importance of features within the input distribution, one may elicit divergent behavior; a greater feature sparsity, for instance, fosters greater superposition. Analysis of the phase diagram of each expert within MoEs establishes that they marshal representational strategies distinct from those of dense models.

Setup. We adhere to the same setup as that of Section 4.1, save for the inclusion of a load-balancing loss to obviate specialization collapse—the eventuality in which experts and routers subside into local minima wherein certain experts are ignored altogether—in certain configurations. Three model setups are considered, each with a single active expert ($k = 1$): (A) $n=2, m=1$; (B) $n=3, m=1$; and (C) $n=3, m=2$. We report the expert-specific phase diagram across feature sparsity and last-feature relative importance for varying network sparsity, obtained by augmenting the experts (E) up to the number of input feature dimensions (n).

In this section we hold the active parameters fixed rather than the total parameters, such that $m_{dense} = km_{moe}$ (the router parameters being set aside). The rationale is to permit comparison within model architectures; were it otherwise, any differences observed might be imputed to architectural changes rather than to the number of active parameters. It so happens that 4.C.1/1 possesses the same number of active parameters as 4.B.X/2. The latter, however, disposes of but a single hidden dimension ($m = 1$) with which to encode the same number of input features ($n = 3$) as the former, which enjoys two hidden dimensions ($m = 2$), thereby rendering it difficult to invoke superposition as a means of understanding specialization.

Results and Takeaways. In every single-expert (dense) case, we observed a pronounced phase change (Figure 4.X.1/1), thereby corroborating the findings of Elhage et al. (2022). Once we augmented the total number of experts, however, the discrete phase changes vanished. Certain experts within MoEs configured with $E = 2$ prove reminiscent of their respective dense counterparts (Figure 4.X.2/2), yet they evince more continuous transitions. In each instance, one expert grew more monosemantic, specializing in the feature of greatest relative importance. Experts dissimilar from the dense cases uniformly register far lower superposition scores (appearing bluer), indicative of more monosemantic representations. This accords with the conclusions of the preceding section—MoEs favor lower superposition scores than do their dense counterparts.

In the $n = 2, m = 1$ setup (Figure 4.A), the dense model declines to represent the last feature whenever feature sparsity is low; the comparable MoE model, by contrast, preserves the last feature to a considerably greater degree, precisely because it commands the requisite capacity. With three input dimensions (Figure 4.B), the MoEs forgo this behavior, inasmuch as the experts are superimposing the remaining two features and no room subsists for the third feature within a single hidden dimension. In contradistinction to the other two cases, for architecture B the hidden dimension endowed with superposition does not suffice, in the

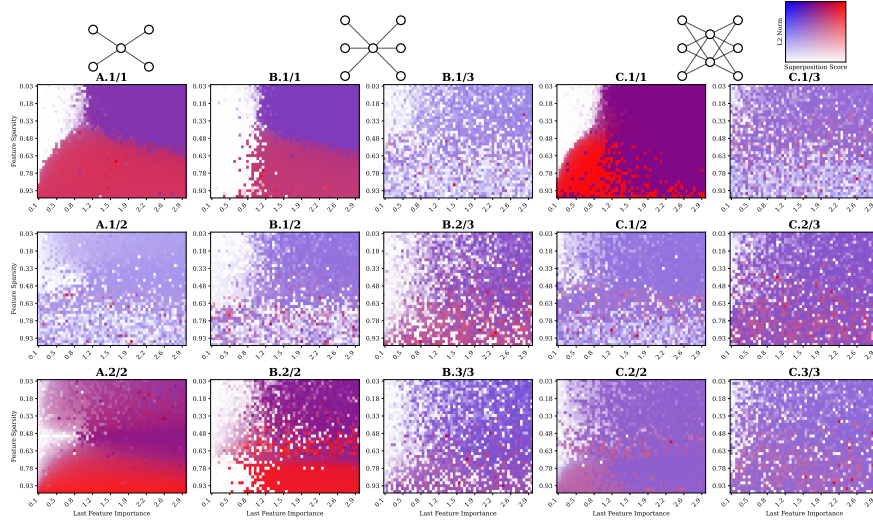


Figure 4: For a given expert and input dimension (feature), we may decode the manner in which it is embedded within the hidden dimension—whether it is ignored (white), rendered monosemantic (blue-purple), or superimposed (red). We plot the joint feature norm ($\|W_n\|^2$) together with the superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) across varying feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, wherein the subscript n designates the last feature among n total features. For each cell, we train ten models and retain the one attaining the lowest loss; a load-balancing loss was employed throughout this section. We plot the joint feature norm and superposition pertaining to the last feature: a low L2 norm ($\|W_n\|$) appears white, signifying that the model disregards the last feature; failing that, a low superposition score appears blue-purple, betokening a monosemantic representation of the last feature, whereas red signals that the feature is represented in superposition. Cell (i, j) within subfigure X.e/E designates expert e of E total experts trained upon architecture X for last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes a dense model.

high-sparsity regime, to represent the full complement of features. We nonetheless discern no clear phase change—the sole exception being the 0.5–0.7 feature-sparsity region in 4.B.2/2, where the pattern is predominantly discrete yet mixed. For $m = 2$, the white region of the dense model (Figure 4.C.1/1)—situated in the mid- to low-feature-sparsity domain, where the feature is of comparatively lesser importance than the others—disregards the last feature. As network sparsity mounts, however, across all remaining panels of Figure 4.C, the models represent the last feature with progressively greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Stated otherwise, the dimensionality within the low-relative-importance region increased in step with rising network sparsity, as demonstrated in Figure 3.

We observed a window of feature sparsity extending from roughly 0.48 to 0.7 in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2, within which a pronounced admixture of polysemanticity, monosemanticity, and outright ignorance obtains. This finding indicates the existence of a middle ground in MoEs—one in which polysemantic and monosemantic representations incur comparable loss—that renders it difficult to commit consistently to the strategies we observe in the low- and high-feature-sparsity domains. We discern no such pattern in dense models, which constitutes evidence that MoEs learn representational strategies of a different order.

Conclusion. These experiments are conducted uniformly under top- $k = 1$, whereby only a single expert is active at any given moment. Even so, we witness markedly divergent behavior in the $E = 2$ case alone—including in those regimes where the hidden-dimension capacity, augmented by superposition, would suffice to represent all features. This leads us to conclude that it is misleading to conceive of MoEs as a mere aggregation of dense

models. The routing mechanism, by permitting experts to observe only a subset of the feature domain, profoundly reshapes both the behavior and the learning of the experts.

6 EXPERT SPECIALIZATION

Given that MoEs exhibit diminished superposition, we now turn to examine the organization of such monosemantic features within experts, together with its bearing upon specialization.

Expert specialization in MoEs has traditionally been construed in terms of load balancing among experts across the totality of inputs (Chaudhari et al., 2025). Such a construal, however, fails to capture the natural intuition of specialization, according to which an expert is enlisted only when the appropriate concepts—those in which the expert is specialized—are present within the input.

We deem an expert specialized if it *occupies* certain feature directions within the input space and if it represents those features in a relatively monosemantic fashion. We demonstrate that these two conditions are directly correlated, and we further exhibit the manner in which their conjoint presence fosters load balancing across experts.

Because we hold $k = 1$ fixed, the feature space is partitioned into convex cone regions (see Appendix A.3), each region being routed to a particular expert. By definition, this entails that $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, wherein C denotes the set of points contained within the cone and s any positive scalar. Should a particular feature vector $\vec{x}$ be routed to an expert, then every $s\vec{x}$ is routed to that selfsame expert. In such a case, we say that feature x is contained within expert e , and that expert e accordingly *occupies* x .

We find empirically that small models which distribute the input space across a greater number of experts tend to attain lower loss (see Appendix A.4). This finding prompts a question: does the allocation of the input space to particular experts betoken any properties of those experts? Our definition of expert specialization suggests that such allocation implies *monosemanticity*—a correlation that, as we shall see, holds even for larger toy models (with $m = 10$, for instance).

In models with $n > 2$, we investigate whether initializing experts so as to occupy features within the input space induces those experts to become more monosemantic with respect to said features. Separately, we ascertain whether, for the features an expert has elected to represent monosemantically, that expert in turn occupies those features within the input space.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied—and that feature alone—as displayed in Figure 5a. When the router is initialized in the ordered k -hot fashion, the first expert monosemantically represents four of the five features it initially occupied, as displayed in Figure 5b; the remaining experts, initialized over other features, neither represented these less important features monosemantically nor represented monosemantically the five most important features over which they were not initialized. When we disrupt the ordering of feature importance and randomize the features each expert initially occupies, each expert came to represent monosemantically only the most important feature or features over which it was initialized, as displayed in Figure 5c.

A strong correlation obtains between the features initially routed to an expert and the features that expert comes to represent monosemantically. We observe, furthermore, that experts represent monosemantically only important features. This holds whether we initialize each expert with a single important feature explicitly or whether we furnish it with a set of features, from which it proceeds to select the most important feature of its own accord and to represent that feature monosemantically.

In the case of uniform feature importance, experts consign all features to superposition with elevated levels of interference. Despite this, the features an expert initially occupies remain *relatively* more monosemantic, attaining on average superposition scores 1.03 standard deviations below the mean for that expert.

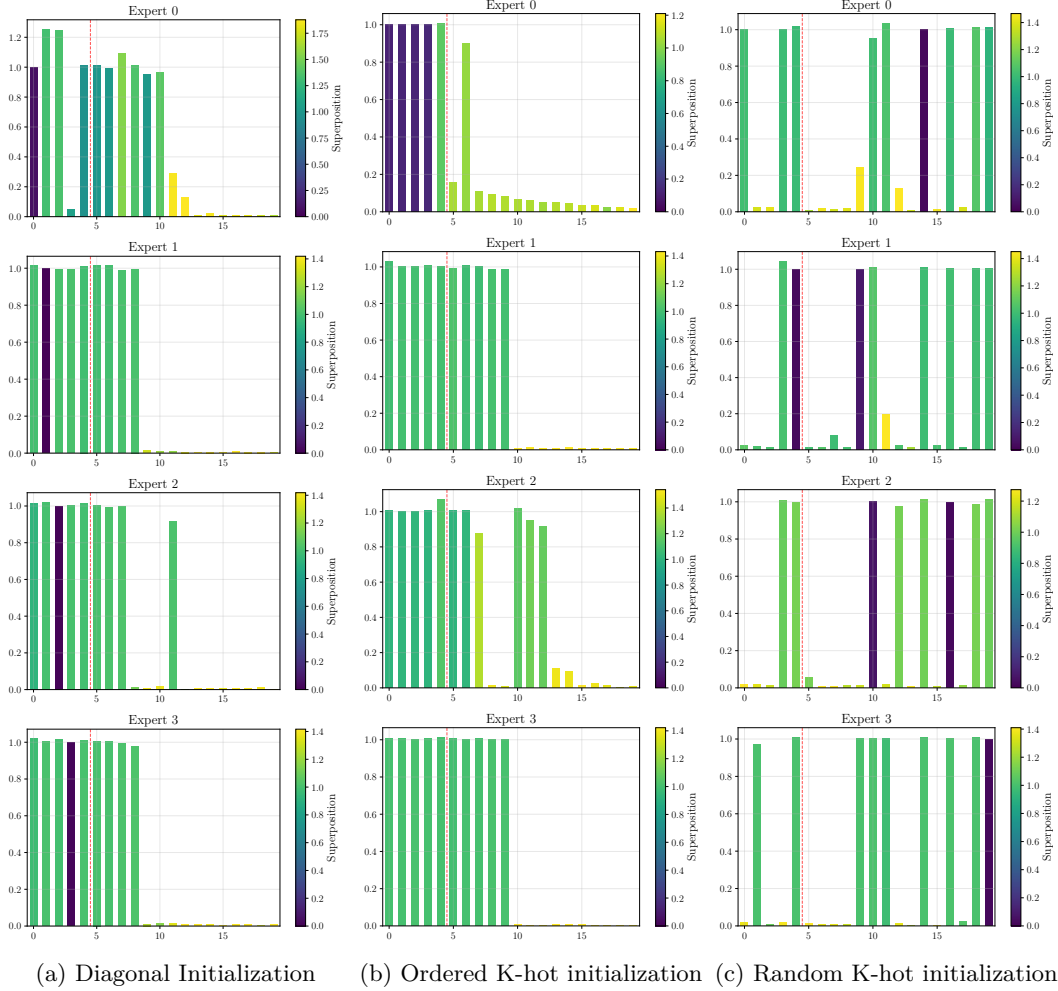


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ together with superposition (rendered as color) for three distinct initialization schemes, under $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), while relative feature importance decays exponentially in sequence from the first feature through the twentieth. In (b), the gate matrix is initialized to an “ordered k-hot,” whereby the first expert aligns with the first five features and each subsequent expert aligns with the ensuing five; relative feature importance is as in (a). In (c), the gate matrix is initialized to a “random k-hot,” wherein each expert is assigned five random features such that no two experts share a common feature yet all twenty features are covered collectively; relative feature importance decays exponentially but is distributed at random across the features.

We investigated whether a correlation obtains between an expert’s representing certain features monosemantically and that same expert’s occupying those features within the input. To this end, we measure usage statistics both when those features are *one among many* active features and when they are the *sole* active features—this latter case being tantamount to measuring the probability that the expert occupies these features. The correlation holds under both the xavier and the k-hot initialization schemes, as may be seen in Table 1. Given $E = 10$, a mean expert usage of $\sim 10\%$ betokens an even load balancing across experts. In every case, whenever the monosemantic feature or features corresponding to an expert are active, the usage of that expert increases appreciably; and whenever such features constitute the sole active feature, the expert dominates the usage entirely. Under the k-hot initialization scheme, *100%* of all features represented monosemantically by an expert are occupied by that selfsame expert.

Table 1: Monosemantic-feature and usage statistics per expert for $n = 100, m = 10, E = 10$. One hundred models are trained under each initialization scheme (xavier and k-hot), yielding 1000 experts apiece. Each statistic is aggregated across models, with experts classified according to the number of features they represent monosemantically. For the feature or features an expert represents monosemantically, we track the expert’s usage both when said feature(s) are one among several active features in the input and when said feature(s) constitute the *sole* active feature in the input.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

As experts come to represent more features monosemantically, they may be regarded as correspondingly more specialized. Their usage on arbitrary input diminishes, yet, conditional upon their specialized features being active, their usage rises far more steeply than that of other experts. This pattern holds across all cases save the xavier-initialized model harboring a four-monosemantic-feature expert, in which a pronounced decline in utilization is observed.

7 CONCLUSION

We have investigated the manner in which experts modulate superposition within MoEs, establishing that MoEs consistently exhibit greater monosemanticity than dense networks while forgoing any phase change. We have proposed a feature-based definition of expert specialization, demonstrating thereby that experts organize themselves spontaneously around coherent features whenever the initialization is contrived to encourage such specialization. Our findings, however, rest upon simple autoencoder toy models furnished with synthetic data, and thus leave unresolved the question of their generalization to large-scale transformers, wherein the feature distribution is unknown (see Appendix A.6). Notwithstanding these limitations, we have shown how toy MoEs attain comparable loss while sustaining more interpretable representations—thereby contesting the prevailing zeitgeist according to which mechanistic interpretability and model capability stand in fundamental tension. Future work ought to interrogate what it is that favors monosemanticity in MoEs, how the training dynamics of MoEs diverge from those of the dense model, and under what conditions specialization emerges. The elucidation of these questions stands to inform the design of language models that are at once more interpretable and more performant.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

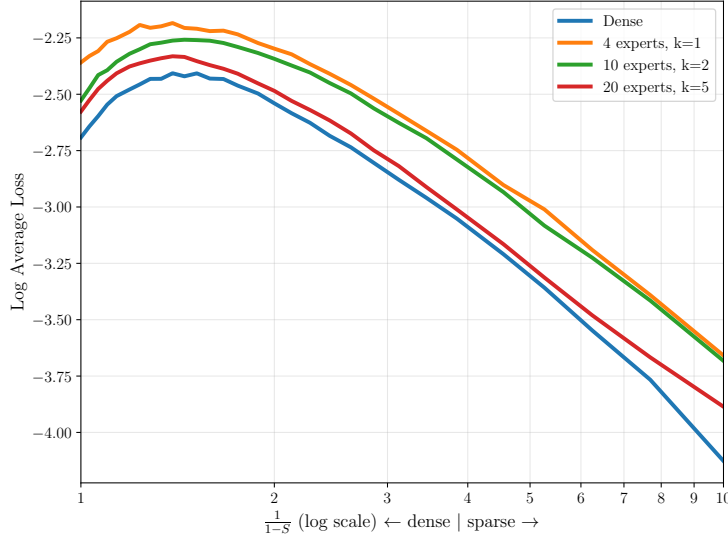


Figure 6: Log average loss plotted against feature density ($\frac{1}{1-s}$) for **dense** ($m = 20$) and **MoE** (4 experts, $k = 1$, $m = 5$), **MoE** (10 experts, $k = 2$, $m = 2$), and **MoE** (20 experts, $k = 5$, $m = 1$) models, all endowed with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. Although the **dense** model outperforms every **MoE** at each sparsity level, the **MoE** loss converges ever more closely upon that of the dense model as the number of experts is augmented.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition arises so as to exploit the discrepancy between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, by contrast, “eat this same gap” structurally by way of conditional computation: through the alignment of activation sparsity with feature sparsity, MoEs abolish the computational penalty ordinarily attached to maintaining dedicated, rarely activating neurons. As Elhage et al. (2022) observe, whenever a model expends computation solely upon active features, the partitioning of polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy, in effect exchanging the *compression* afforded by superposition for the *selection* afforded by routing.

This structural transformation manifests itself geometrically as a reduction in *interference*. Although the global ratio of the number of features represented to the number of parameters remains invariant, the router effectively partitions the feature space, thereby ensuring that a feature routed to expert e contends for capacity only with the subset of features likewise assigned to that expert. In consequence, interference comes to be governed by the local expert matrix $(W^e)^\top W^e$ rather than by the global $W^\top W$ of a dense model. Such partitioning drastically curtails the number of interfering features for any given feature vector, thereby minimizing the optimization pressure to pack features in superposition and enabling experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime of $k = 1$, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$. The region routed to expert i admits of representation as $\forall j \neq i, (w_i - w_j)^\top x > 0$, wherein w_i and w_j denote row vectors of W^r . This constitutes a homogeneous

linear inequality, and regions delimited by such inequalities are, by definition, convex cones. Should a particular x satisfy this inequality, then multiplication of both sides by any positive scalar s leaves it satisfied; and should x_1 and x_2 both satisfy the inequality, then so too does any $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$.

In the case of $k > 1$, the region of inputs consigned to a particular expert e becomes a union of convex cones. In general, however, the union of convex cones is not itself a convex cone; the conception of experts as occupying feature directions may therefore fail to hold beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

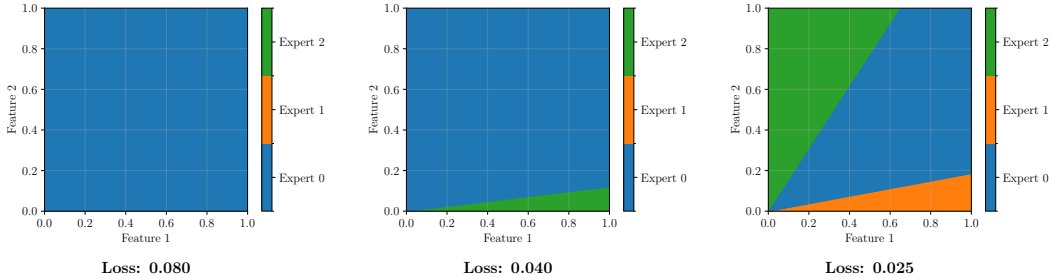


Figure 7: Expert routing across three identical models distinguished by their initialization schemes; we employ $n = 2$, $E = 3$, $m = 1$. The first model (left) attains the poorest performance (loss: 0.08) and consigns all inputs to a single expert. The second model (middle) attains superior performance (loss: 0.04) and consigns a small portion of inputs—specifically those for which feature 1 is active—to a second expert. The third model (right) attains the lowest loss (loss: 0.025) and distributes the input space among all experts: one expert is elected when feature 1 alone is active, one when feature 2 alone is active, and one when both are active.

In small models ($n = 2, m = 1, E > 1$), we find empirically that models distributing the input space across a greater number of experts tend to attain lower loss, as borne out by testing with $E \in [2, 7]$. Holding $n = 2$ fixed affords us the opportunity to visualize which portions of the input space are routed to which experts, as displayed in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map between any dense model and a monosemantic MoE possessing an equivalent number of active features under a sparsity constraint.

Assume there to exist an upper bound a upon the number of active features for any input, such that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Assume, moreover, that a is no greater than the hidden dimensionality m of an expert, which furnishes an upper bound upon the number of features a model must represent; and assume likewise that the hidden dimensionality is smaller than the total number of input features n (whence $a \leq m \leq n$). To construct the monosemantic MoE, for each admissible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —that is, each subset of active features whose cardinality does not exceed a —we create an expert that monosemantically preserves those features. (One may in fact restrict attention to those subsets for which $|S| = a$.) The router thereupon selects the expert corresponding to the active features in question, of which, by assumption, there will never be more than a :

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

wherein $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert commands sufficient capacity to represent its assigned features without recourse to superposition. To reiterate,

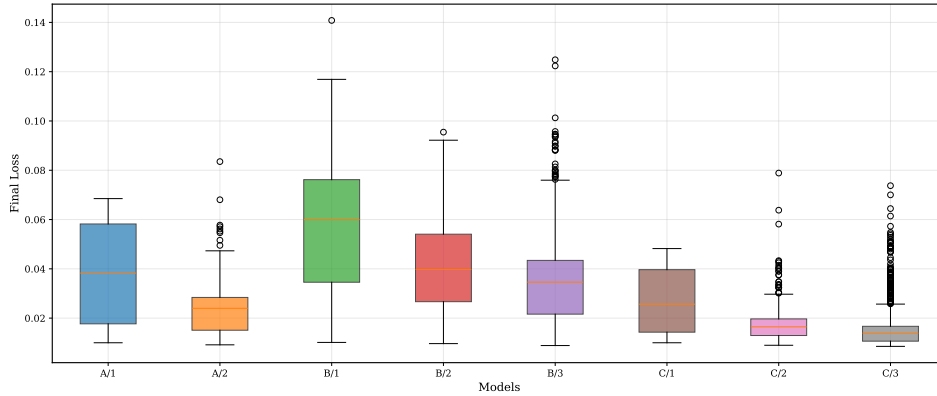


Figure 8: In the notation X/E , X denotes the same model architectures and models employed in Figure 4, while E denotes the total number of experts (that is, the network sparsity). Increasing network sparsity diminishes the mean loss even as it amplifies localized variance—the more markedly so as the number of experts approaches the input feature dimensions. This phenomenon may be imputed to the comparatively unstable training of MoEs relative to dense models (notwithstanding the training of ten models per cell and the selection of the lowest loss).

only a features are ever active, and every distinct combination of active features is apportioned its own dedicated expert possessing sufficient capacity to represent those features monosemantically. Consequently, the number of experts that may be required is $\binom{n}{m}$.

The reconstruction effected by this theoretical MoE incurs zero loss only insofar as the sparsity constraint holds (or, in these toy models, tends toward unity), for there subsists the possibility that more than m features be active at once ($a \not\leq m$), an eventuality that would exceed the monosemantic representational capacity of the network—though the dense polysemantic model could fare no better, save where features are correlated within the distribution. Hence, even when $a \not\leq m$ on occasion, the polysemantic model is beset by the same difficulty, and the monosemantic MoE furnished by this construction may yet outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ subject to the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings expounded in this paper, the sparsity constraint holds in the limit as sparsity approaches unity. In practice, however, there may obtain an upper bound upon the number of features that a given quantum of information can semantically encode, as indicated by the size of meaningful embeddings of that data. A MoE model endowed with sufficient experts and a tractable—which is to say interpretable—measure of superposition may accordingly suffice to encode all features present.

A.6 LIMITATIONS

Our results warrant interpretation in light of several limitations. First, all experiments are conducted within controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). Although this affords precise measurement of interference and monosemanticity, it abstracts away many of the complexities attending large-scale transformers—multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics among them. In consequence, the transferability of our findings to real-world MoE architectures remains uncertain.

Second, our architectural and routing choices are deliberately circumscribed: we employ simple experts, fixed top- k routing (frequently $k = 1$), equal parameter budgets across dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs, by contrast, often deploy richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or extend beyond the task of reconstruction to that

of next-token prediction, any of which may engender qualitatively different representational behaviors.

Third, the feature distributions instantiated in our synthetic tasks—encompassing sparsity patterns, feature importance, and independence assumptions—are appreciably simpler than those encountered in natural data.

Finally, although we observe heightened monosemanticity and attenuated superposition in MoEs under fixed conditions, we refrain from evaluating downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work intimates that such dynamics may shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Taken in the aggregate, our study furnishes mechanistic insights under clean experimental conditions, yet further work is requisite to validate these patterns within large, realistic MoE systems. Regrettably, our methods do not scale naturally to larger models, and this constitutes a manifest direction for future work within this space.